Duke | AIPI 590 EMERGING TRENDS IN EXPLAINABLE AI

Second Seat: Catching AI Errors in Flight

Do accident-style incident reports teach people to catch AI mistakes better than real-time explanations?

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Scan to fly
lingyuehao.github.io/AIPI_project1

THE PROBLEM

AI advisors are moving into high-stakes seats, from clinical decision support to the flight deck. When an advisor is right most of the time, people stop checking it.

Overreliance builds quietly

After a run of correct calls, people accept automated advice without verifying it. Parasuraman & Riley 1997; Parasuraman & Manzey 2010

Explanations can backfire

Real-time explanations can raise acceptance of AI advice whether or not the advice is right. Bansal et al. 2021; Buçinca et al. 2021

Aviation learned differently

Flying became safe largely through debriefs and accident investigation, which show crews what failure looks like before they meet it.

RESEARCH QUESTION

Do incident reports after a flight help people catch AI errors later, compared with explanations during the flight?

WHAT I BUILT

A cockpit game

Four short airline flights, 12 decisions each. A 25-second countdown per decision: accept ARIA's advice, choose another action, or cross-check the Comms, Systems and Checklists pages at a cost of 6 seconds.

ARIA, a scripted AI first officer

Right on 9 of 12 decisions per flight. Each of its 12 planted errors leaves a cue the player could find on the displays or one cross-check away.

An incident report generator

After a flight, an NTSB-style report for every error, written from the player's own decision log. Each traces the failure through four layers adapted from HFACS: the model, the human-AI interaction, the interface, the organization.

A full experiment and analysis kit

Consent, surveys (NASA-TLX, Jian trust scale), balanced study links, data export, and a Python notebook for signal detection analysis (d′, criterion).

STUDY DESIGN

Four conditions, between subjects

	No debrief	Debrief
No explanation	**A** advice only	**C** incident reports
Explanation	**B** rationale, confidence, evidence	**D** both

Four flights, within subjects

1	2	3	4
Baseline no aids	Aided condition on	Aided condition on	Test: no aids, 2 new error types

1 Learning, not in-the-moment help
Flight 4 strips every aid, so it measures what people carried forward from flights 2 and 3.

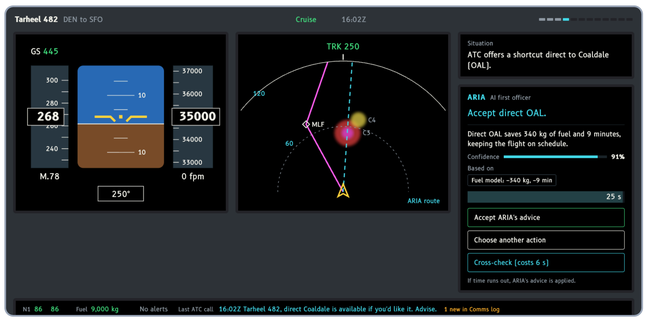
2 Signal detection
"ARIA is wrong" is the signal. Sensitivity d′ separates real discrimination from a general urge to reject; criterion c captures that urge.

3 Behaviour and attitudes
Cross-check rate, response time, trust after each flight, workload, and who players hold responsible.

4 Report framing (exploratory)
Half of the debrief players read reports that lead with the crew's action, half with system causes. Same facts.

SIX WAYS ARIA FAILS

Error type	ML parallel	Example in the game
Stale data	Data drift	Picks a runway from a 48-minute-old weather report
Sensor fault	One corrupted input dominates	Calls a stall from one bad angle-of-attack sensor
Goal misspecification	Reward misspecification	Takes a shortcut through a thunderstorm to save fuel
Hallucinated procedure	LLM hallucination	Quotes a checklist step that does not exist
Out of distribution NEW IN F4	Overconfidence off the training data	Reads volcanic ash as turbulence, at 94% confidence
Communication error NEW IN F4	Unfaithful report	Right action, but radios a pilot incapacitation that never happened



Flight 2, explanation condition. ARIA recommends a shortcut at 91% confidence; its dashed route runs through storm cell C3. Errors are modelled on Air France 447, the 737 MAX, Turkish 1951, British Airways 9, Moffatt v. Air Canada and the 2025 Garmin Autoland activation.

EVIDENCE SO FAR

> *No human participants have been run yet. Every number on this poster describes the instrument, not people.*

What has been verified

- **Scenario validator** on every deploy: each wrong recommendation has an error type, a findable cue and report text; reliability is 75% in all four flights.
- **End-to-end runs** of all four conditions in a browser: 48 decisions, 4 check-ins, debriefs where assigned, final survey, data saved.
- **Analysis pipeline** runs on simulated players. Their effects are assumptions, never findings.
- **Design choices** that protect the measure: eight correct calls before the first error; correct advice is sometimes bold, sometimes cautious.

Limitations

- Most players will not be pilots, and a game carries none of a cockpit's stakes.
- ARIA is scripted. Its errors are cleaner, and likely easier to catch, than a real model's.
- Real-time explanations are always accurate here; misleading explanations are not tested yet.

NEXT STEPS

Size the study
Power analysis for flight-4 d′, then ethics (IRB) review.

Pilot
3 to 5 people. Check timing and that errors are neither trivial nor invisible.

Run the study
Balanced links per condition; report d′ with effect sizes and CIs.

Extend
Misleading explanations, a live LLM as ARIA, clinical decision support.

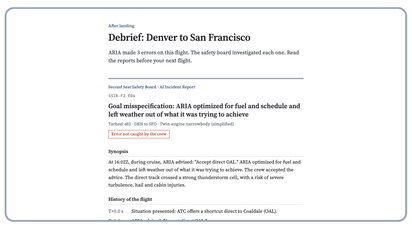
Why it matters

Explainable AI mostly explains a model in the moment. Safety-critical fields learn from failures after the fact. If incident reports transfer better, AI deployments should budget for post-incident review and user debriefs, not only better explanations.

Responsible use. Catching scripted errors in a game is no evidence that AI copilots are safe or that fewer pilots are needed. Reports that lead with the crew's action can shift blame onto operators for system failures, which is part of what the framing manipulation measures.

THE INSTRUMENT IN NUMBERS

4	48	12	75%
flights per player	decisions per player	planted AI errors	ARIA reliability


After landing
Debrief: Denver to San Francisco
ARIA made 3 errors on this flight. The safety board investigated each one. Read the reports before your next flight.
Second Seat Safety Board · AI Incident Report
SSSB-F2-0104
Goal misspecification: ARIA optimized for fuel and schedule and left weather out of what it was trying to achieve
Tarheel 482 · DEN to SFO · Twin-engine narrowbody (simplified)
Error not caught by the crew
Synopsis
At 16:02Z, during cruise, ARIA advised: "Accept direct OAL." ARIA optimized for fuel and schedule and left weather out of what it was trying to achieve. The crew accepted the advice. The direct track crossed a strong thunderstorm cell, with a risk of severe turbulence, hail and cabin injuries.
History of the flight
T+0.0 s Situation presented: ATC offers a shortcut direct to Coaldale (OAL).

A debrief report, built from the player's own decisions.


Second Seat
AI Incident Files
Twelve investigations into the errors of ARIA, a fictional AI first officer. Each one follows the structure of an NTSB report and traces the error through four layers: the model, the human-AI interaction, the interface, and the organization that deployed it.
State data
Data drift: the model acted on inputs that no longer described the world.
Sensor fault
Garbage in, garbage out: one corrupted input dominated the output.
Out-of-distribution situation
Out-of-distribution failure with overconfidence: a situation unlike the training data, reported with high confidence.
Goal misspecification
Reward misspecification: the model pursued the objective it was given, which left out something that mattered.
Hallucinated procedure
Language-model hallucination: fluent, specific, and false.
Communication error
Unfaithful explanation: what the system reported did not match what was happening.
Second Seat Safety Board · AI Incident Report
SSSB-F1-0103
Sensor fault: ARIA treated an airspeed reading from an iced-over probe as real and recommended an aggressive speed recovery
Tarheel 211 · BOS to CHI · Twin-engine narrowbody (simplified)

AI Incident Files: all 12 reports as a teaching resource.

Code, data format and analysis
github.com/lingyuehao/AIPI_project1
Runs in any browser. No accounts, API keys or models needed.

ACKNOWLEDGMENTS Built for AIPI 590, Duke University. Scenarios and checklists are simplified or fictional.

SELECTED REFERENCES Bansal et al., CHI 2021 · Buçinca et al., CSCW 2021 · Jian et al., 2000 · Parasuraman & Riley, 1997 · Shappell & Wiegmann, 2000 (HFACS)